

Unequal Job Opportunities and the Measurement of Welfare Inequality

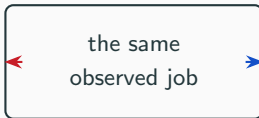
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LISER and University of Luxembourg

France 2016 · EU-SILC priced through EUROMOD

The same job can mean opposite things

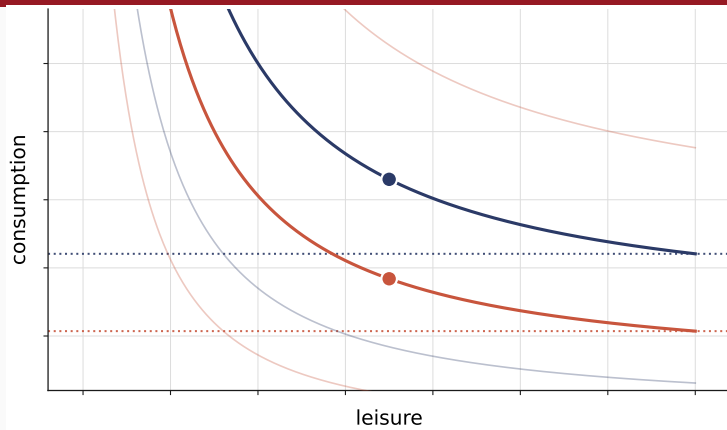
one person
chose it
from a wide set



the other
had almost
nothing else

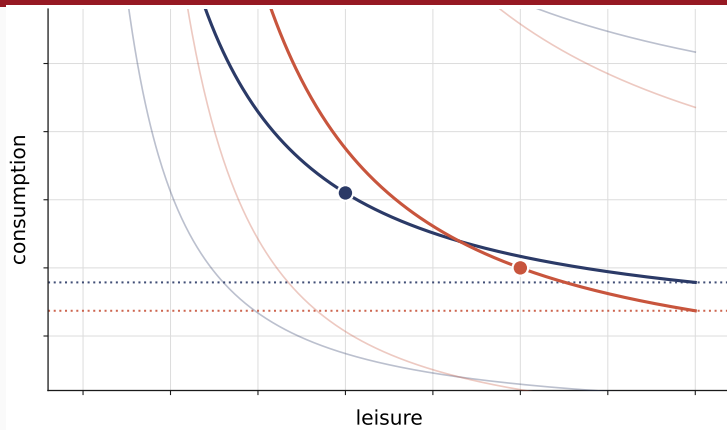
The taste parameter absorbs whatever the choice set cannot express.

Preferences and opportunities are separate objects



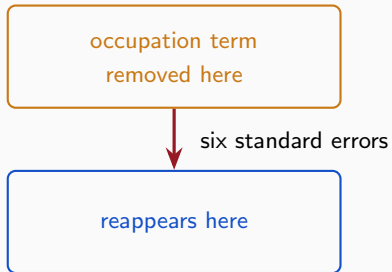
Same preferences, different opportunities.

Preferences and opportunities are separate objects



Same opportunities, different preferences.

Drop the opportunity object and the effect relocates, it does not vanish



The variation loads on whichever channel is left open.

Availability is four estimated factors, jointly with preferences

$$g = \underbrace{g^E} \underbrace{g^H} \underbrace{g^{\text{Occ}}} \underbrace{g^W}$$

whether work at what in which at what
is available hours occupation pay

The hours factor carries no household-specific coordinate.

Each of 1,555 households gets its observed job plus 100 drawn ones

$$V_{ij} = u_{ij} + \log g_{ij} - \log q_{ij}$$

preferences opportunity sampling correction

q is computation. g is economics.

Every job is priced.

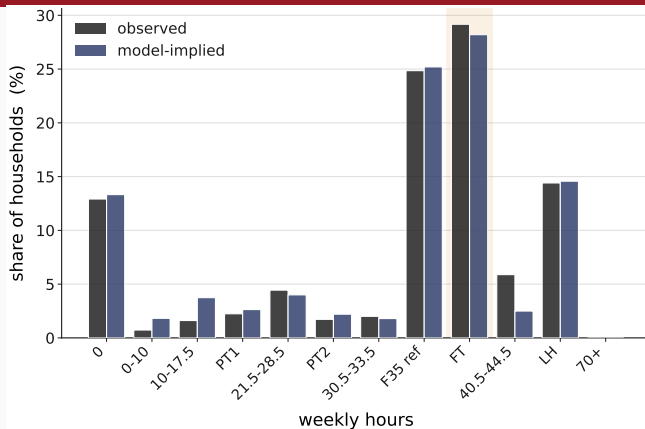
Two things stay conditional, and both are stated as limits

access *versus* capability

persistent household heterogeneity

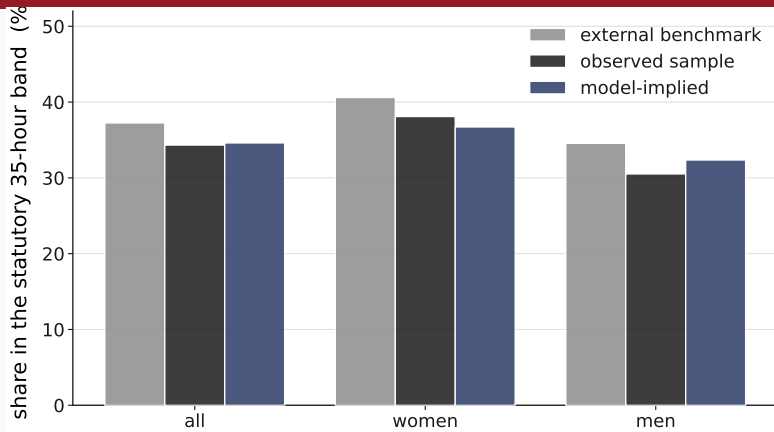
Structural and model-conditional. Not causal.

The model reproduces the statutory week: 25.20% against 24.85% observed



An opportunity peak in the offer density. Not a causal estimate.

An outside source the model never saw puts the statutory band at 37.2%



Validation only. No external moment enters the likelihood.

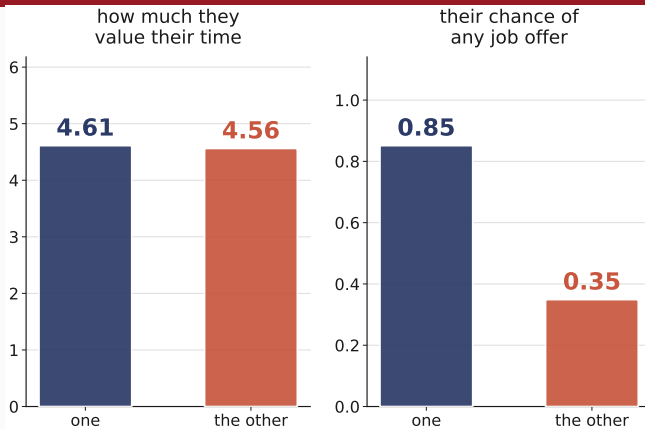
Welfare is the flat pay that would leave you as well off

$$W_i = W^1(z_i, R_i, g_i; \mathbf{y}^w)$$

	own preferences	reference preferences
own the environment	l_{00}	l_{10}
reference the environment	l_{01}	l_{11}

Pay is compensated; the opportunity set is not.

Welfare is the flat pay that would leave you as well off



Two real households, the same preferences, different opportunity.

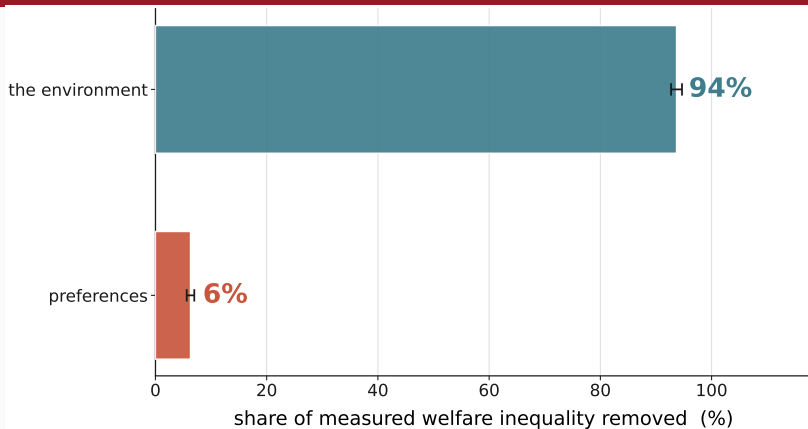
The environment carries 94% of measured welfare inequality

$$\underbrace{C_{\text{pref}}}_{\text{preferences}} + \underbrace{C_{\text{env}}}_{\text{the environment}} = I_{00} - I_{11}$$

$$C_{\text{env}} = \underbrace{C_{\text{acc}}}_{\text{job access}} + \underbrace{C_{\text{earn}}}_{\text{earning opportunities}} + \underbrace{C_{\text{needs}}}_{\text{endowments and needs}}$$

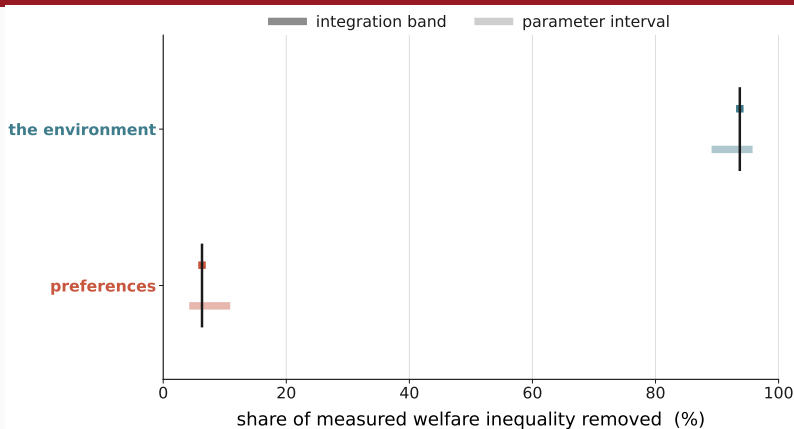
The fully common state is numerically zero — a tested property.

The environment carries 94% of measured welfare inequality



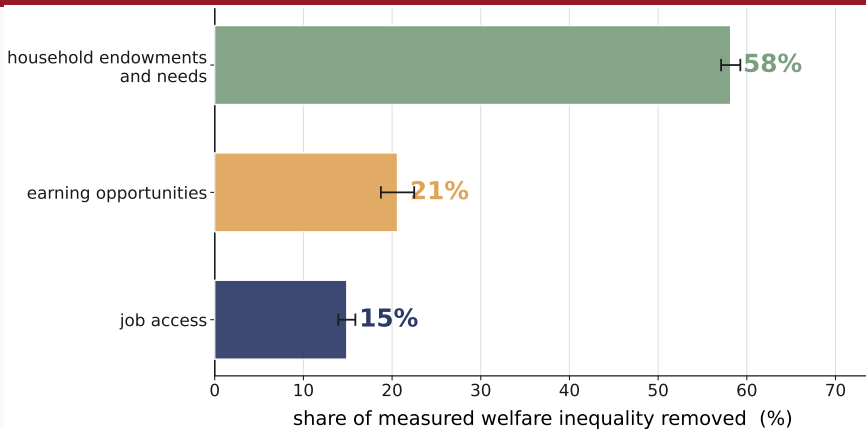
Bars are integration precision, not confidence intervals.

The environment carries 94% of measured welfare inequality



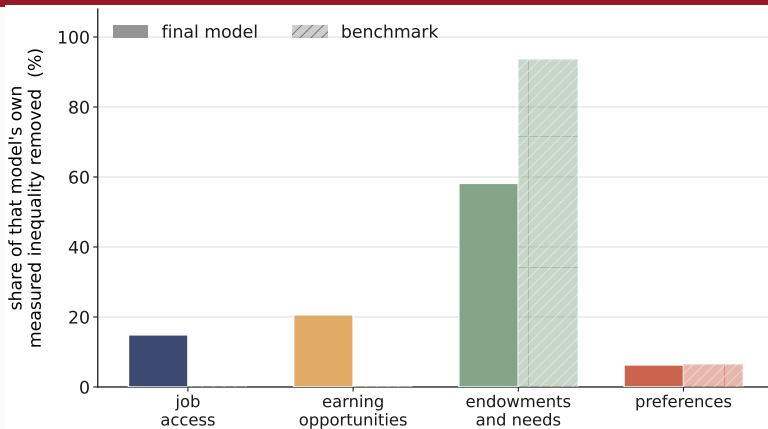
Two kinds of interval, on offset rows. Never merged.

Inside it, the budget side carries 58% and the market side 36%



No ordering is claimed between the two market channels.

Omit the opportunity object and measured inequality falls 24%



The missing share is relabelled, not turned into taste.

The preference share is the fragile quantity: 6% to 11%

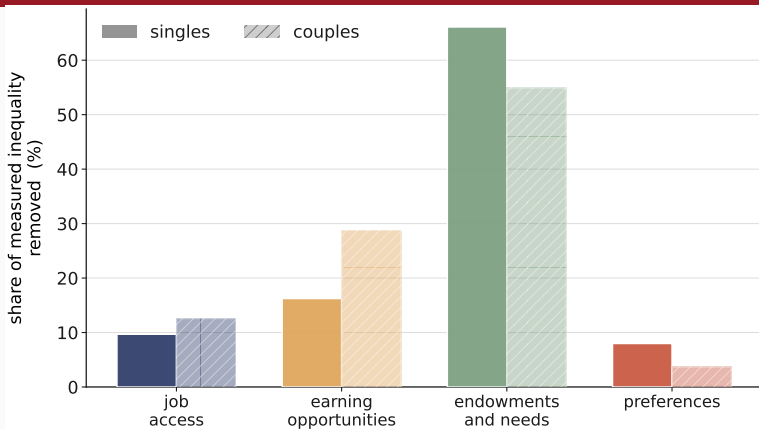
preferences ●—————● moves with the reference

the environment ●—●

barely moves

No sign and no ordering changes anywhere.

Couples reproduce the same order, with a sensitive preference share



Singles remain the quantitative headline.

Separating wants from opportunities changes the attribution

Method every job priced, opportunities estimated

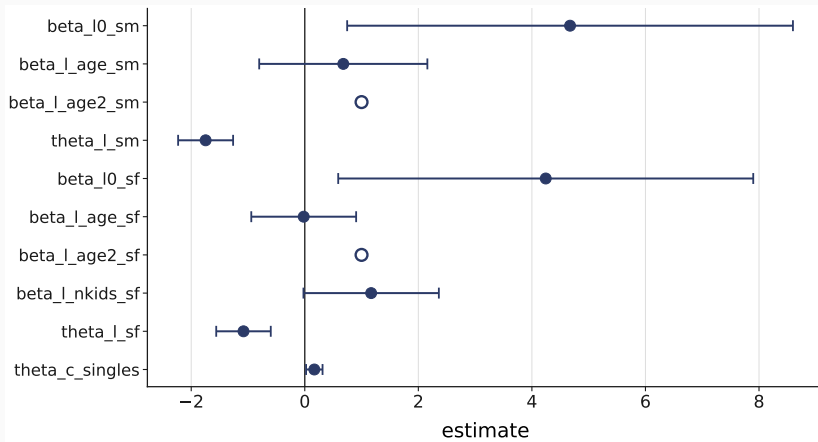
Finding the environment carries 94%

Margin preferences 6% to 11%

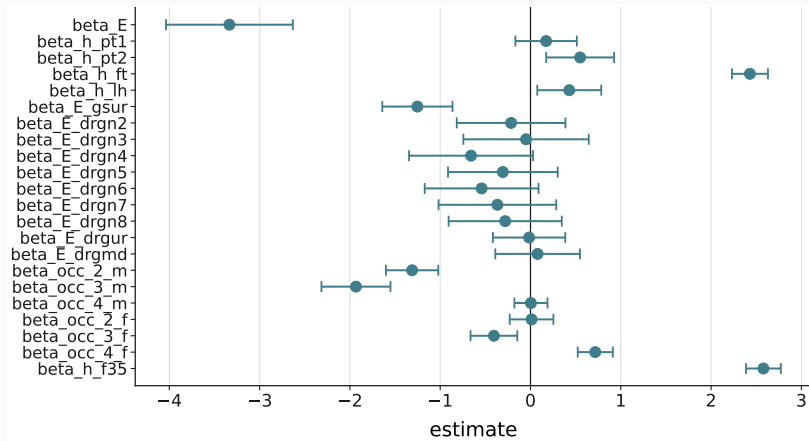
$$u_{ij} = \beta_{\ell}(\mathbf{x}_i) \underset{\text{leisure}}{\text{BC}(\ell_{ij}; \theta_{\ell})} + \underset{\text{consumption}}{\text{BC}(c_{ij}; \theta_c)}$$

Box–Cox in both; consumption normalised.

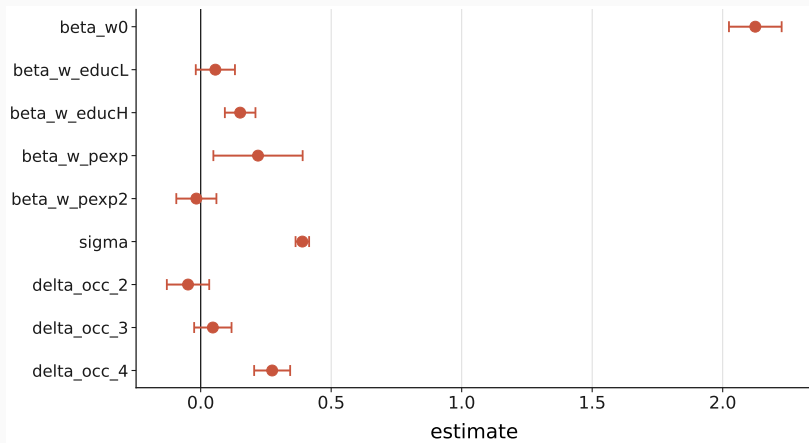
B1 — The estimated coefficients: preferences



B1 — The estimated coefficients: job access



B1 — The estimated coefficients: earning opportunities



B2 — Exhaustiveness is a tested property

defect	what fixed it
a missing budget channel	a fourth player from its own marginals
asymmetric money-metric inversion	the reference side moves with the coalition
an incomplete male reference block	a structural zero on the child term

The fully common state is 0.000000 — numerically zero.

B3 — Equivalization moves the scale with the needs channel

	raw	equivalized
preferences	6.30%	7.98%
the environment	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing the own scale leaves exactly the Gini of its inverse.

B4 — The hours-wish evidence has no model counterpart

ground	what it shows
definitional	the candidate's optimum tracks pay
identification	the hours block moves the moment by exactly 0
numerical	the level can agree; the profile cannot

Tried, and it fails on three independent grounds.

B5 — Where the welfare family comes from

Compensation

unequal productivity should not become
unequal well-being

Responsibility

preference-based choices should not be
compensated

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet, SCW 2005]

W^1 is one compromise — the one this paper takes to the data.

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).